

Midterm Exam (Solution Hints)

(6:35 PM – 7:50 PM : 75 Minutes)

- This exam will account for 25% of your overall grade.
- There are four (4) questions, worth 75 points in total. Please answer all of them in the spaces provided.
- There are 14 pages including seven (7) blank pages. Please use the blank pages if you need additional space for your answers.
- This is a *closed book, closed notes* exam. *No cheat sheets* are allowed.
- You are *not allowed to use calculators*.

Good Luck!

Question	Parts	Points
1. Recurrences	$(a)-(j)$	$2.5 \times 10 = 25$
2. Complexity Analysis	$(a), (b), (c)$	$7 + 7 + 6 = 20$
3. Partition	–	15
4. Bucket Sort	$(a), (b), (c)$	$5 + 5 + 5 = 15$
Total		75

NAME: _____

SBU ID: _____

Question 1. [15 Points] Recurrences. For each of the functions below either solve it using the Master Theorem (and state the case that applies) or state why the Master Theorem does not apply. Assume that $T(n) = \Theta(1)$ for $n \leq 1$. All logarithms have base 2.

(a) [2.5 Points] $T(n) = 4 \cdot T\left(\frac{n}{\log n}\right) + n^2$

Master theorem cannot be applied as $\mathcal{B}$ is not a constant.

(b) [2.5 Points] $T(n) = 2 \cdot T\left(\frac{n}{2}\right) + \sin n + 5$

$T(n) \in \Theta(n)$ because $(\sin n + 5) \in \mathcal{O}(1)$ and using case 1 of the theorem.

(c) [2.5 Points] $3T(n) = T\left(\frac{n}{2}\right) + T(n) + 3T\left(\frac{n}{2}\right) + \log^2 n$

$T(n) = 2T\left(\frac{n}{2}\right) + \frac{1}{2} \log^2 n$ $T(n) = \Theta(n)$ using Master theorem case 1.

(d) [2.5 Points] $T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + \log n$

Master theorem cannot be applied as the recurrence is not in the required format.

(e) [2.5 Points] $T(n) = \sqrt{2}T\left(\frac{n}{\sqrt{3}}\right) + n \log \frac{1}{4}$

Master theorem cannot be applied when $\mathcal{F}(n)$ is a negative function (since $\log \frac{1}{4}$ is negative).

(f) [2.5 Points] $T(3n) = 4n + T(2n)$

Replace n with $n/3$ to get $T(n) = T\left(\frac{n}{3/2}\right) + 4n/3$

$T(n) = \Theta(n)$ using case 3 of the theorem

Question 2. [20 Points] Complexity Analysis. For each of the programs given below, write a recurrence describing its running time. Either solve the recurrence using the Master Theorem (and state the case that applies) or state why the Master Theorem does not apply.

(a) [7 Points] The FINDMINMAX program shown in Figure 1.

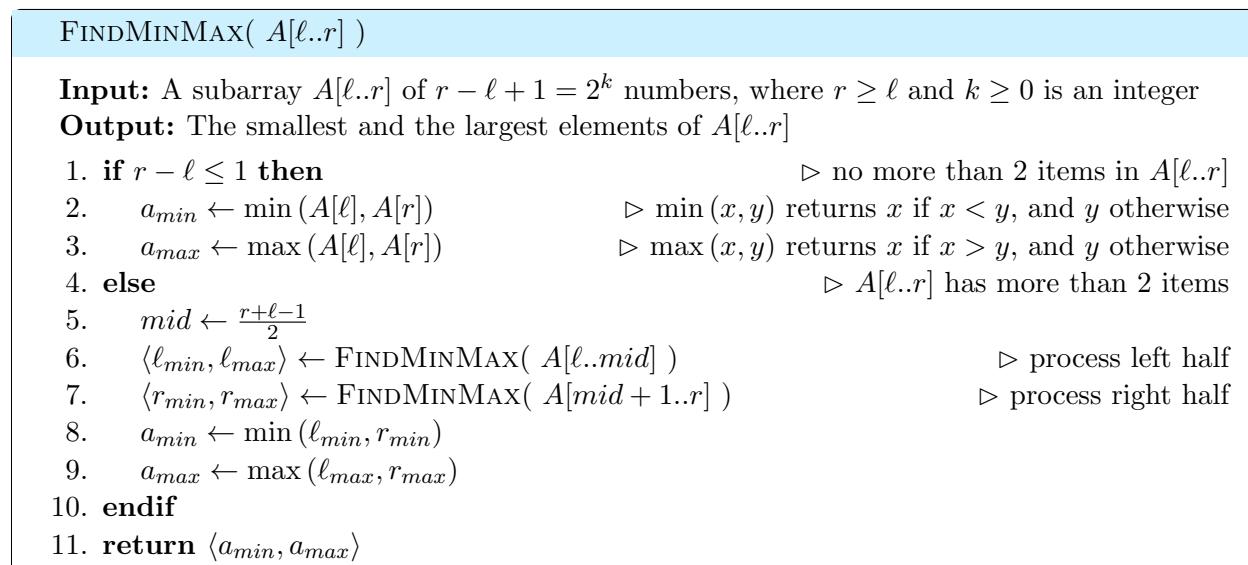


Figure 1: The FINDMINMAX program.

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + O(1)$$

$T(n) \in \Theta(n)$ using case 1 of the theorem.

(b) [7 Points] The STOOGESORT program shown in Figure 2.

STOOGESORT($A[0..n-1]$)	
Input: An array $A[0..n-1]$ of n numbers	
Output: The items in $A[0..n-1]$ sorted in nondecreasing order of value	
1. if $n = 2$ then	
2. $x \leftarrow \min(A[0], A[1]), y \leftarrow \max(A[0], A[1])$	
3. $A[0] \leftarrow x, A[1] \leftarrow y$	
4. elseif $n > 2$ then	
5. $m \leftarrow \lceil \frac{2n}{3} \rceil$	
6. STOOGESORT($A[0..m-1]$)	$\triangleright A[0..m-1]$ has approximately $\frac{2n}{3}$ items
7. STOOGESORT($A[n-m..n-1]$)	$\triangleright A[n-m..n-1]$ has approximately $\frac{2n}{3}$ items
8. STOOGESORT($A[0..m-1]$)	
9. endif	

Figure 2: The STOOGESORT algorithm.

$$T(n) = 3 \cdot T\left(\frac{2n}{3}\right) + O(1)$$

$$T(n) \in \Theta\left(n^{\log_{3/2} 3}\right) \text{ using case 1 of the theorem.}$$

(c) [**6 Points**] The MYSTERY program shown in Figure 3.

MYSTERY(n)
Input: Positive integer n Output: None 1. if $n = 1$ then return 2. if n is even then 3. MYSTERY($n/2$) 4. else 5. MYSTERY($3 \times n + 1$) 6. endif

Figure 3: The MYSTERY program.

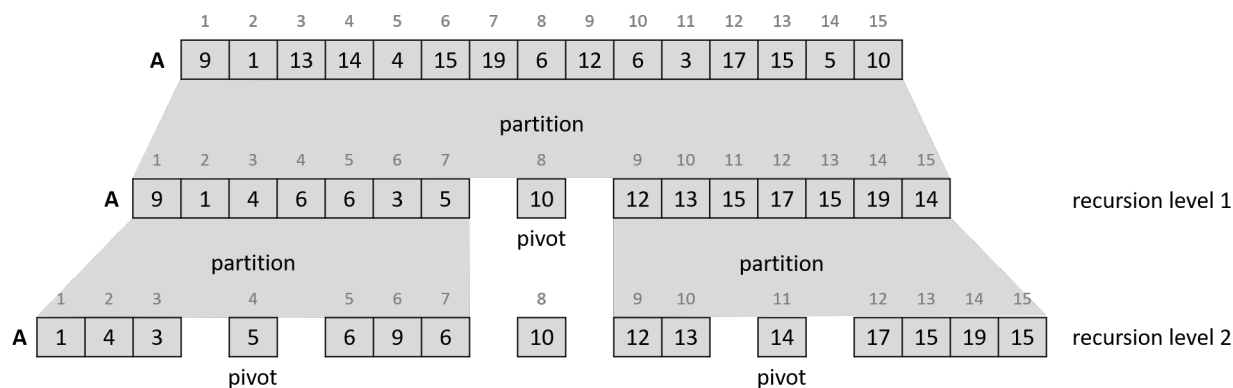
$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ T\left(\frac{n}{2}\right), & \text{if } n \text{ is even} \\ T(3 \times n + 1), & \text{if } n \text{ is odd} \end{cases}$$

Master theorem cannot be applied as a and b are not fixed.

Question 3. [15 Points] Partition.

Recall the in-place partitioning algorithm we learnt in the class (before midterm) which is used by quicksort to partition the numbers in the array or subarray it is working on around the pivot. We always chose the last element of the given array/subarray as the pivot.

Given an array $A[1 : 15]$ as an input the following example shows the state of A after partitioning in the first two levels of recursion of quicksort.



You can describe the state of A after partitioning in recursion level 1 as:

$[9, 1, 4, 6, 6, 3, 5] \ 10 \ [12, 13, 15, 17, 15, 19, 14]$

Then after partitioning in recursion level 2 as:

$[[1, 4, 3] \ 5 \ [6, 9, 6]] \ 10 \ [[12, 13] \ 14 \ [17, 15, 19, 15]]$

Now, given the following array $A[1 : 14]$ of numbers describe its state after partitioning in recursion levels 1, 2 and 3.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	27	2	7	18	31	40	16	12	5	23	12	37	11	17

Level 1: $[2, 7, 16, 12, 5, 12, 11] \ 17 \ [31, 23, 40, 37, 27, 18]$

Level 2: $[[2, 7, 5] \ 11 \ [16, 12, 12]] \ 17 \ [18 \ [23, 40, 37, 27, 31]]$

Level 3: $[[[2] \ 5 \ [7]] \ 11 \ [[12] \ 12 \ [16]]] \ 17 \ [18 \ [23, 27], 31, [40, 37]]$

Question 4. [10 Points] Bucket Sort

The BUCKETSORT algorithm shown in Figure 4 can be used to sort an array of numbers $A[1..n]$, where $0 \leq A[i] < 1$ for $1 \leq i \leq n$.

Answer the following questions for the BUCKETSORT algorithm:

- (a) [5 Points] Find a best-case input for the algorithm. Give its best-case time complexity.
- (b) [5 Points] Find a worst-case input for the algorithm. Give its worst-case time complexity.
- (c) [5 Points] What modification to the algorithm can change the worst-case running time to $\Theta(n \log n)$?

BUCKETSORT($A[1..n]$)
Input: An array $A[1..n]$ of n numbers, where $0 \leq A[i] < 1$ for $1 \leq i \leq n$ Output: The numbers in $A[1..n]$ sorted in nondecreasing order of value
1. create n empty lists $B[0], B[1], B[2], \dots, B[n-1]$
2. for $i \leftarrow 1$ to n do
3. $j \leftarrow \lfloor n \times A[i] \rfloor$
4. insert $A[i]$ into list $B[j]$
5. for $i \leftarrow 0$ to $n-1$ do
6. sort list $B[i]$ with insertion sort
7. concatenate the lists $B[0], B[1], B[2], \dots, B[n-1]$ in order

Figure 4: The BUCKETSORT algorithm.

- (a) The best case will be when every number will fall into a different bucket. For example, a list where the i -th number in the sorted order is $\frac{i-1}{n}$. In this case, the complexity will be $\Theta(n)$.
- (b) The worst case will be when every number will fall into the same bucket. For example, a list where all the numbers are in decreasing order and given is such a way that $\lfloor n \times A[i] \rfloor$ yields the same bucket id. In this case, the complexity will be $\Theta(n^2)$.
- (c) Instead of using insertion sort in line 6, we can use merge sort to change the worst-case running time to $\Theta(n \log n)$

Question 5. [10 Points] Asymptotic Bounds. For each of the following statements state whether it is *True* or *False*. You must justify each answer. Assume that all log's are of base 2.

(a) [5 Points] $n^{\frac{1}{\log n}} = \Theta\left((\log n)^{\frac{1}{\log \log n}}\right)$

(b) [5 Points] $(2^{\sin 2^n}) \log n = \mathcal{O}(\sqrt{n})$

(a) True

In general, for any term x , $x^{\frac{1}{\log x}} = (2^{\log x})^{\frac{1}{\log x}} = 2 = \Theta(1)$. For left hand side, we can take $x = n$ and get $\Theta(1)$. For right hand side, we can take $x = \log n$ and get $\Theta(1)$.

(b) True

$2^{\sin 2^n}$ will be between $\frac{1}{2}$ and 2. So it is a constant. $\log n < n$ holds true for all $n > 0$.
So, $\log n = 2 \log \sqrt{n} \leq \sqrt{n}$